# Lecture 19

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# Unification for FOL

- Let $C$ be a clause; if we negate all literals in $C$, we get $C^-$
- A unifier for a clause $C=\{l_1,\ldots,l_n\}$ is a unifier for $\{(l_1,l_2), (l_2, l_3), \ldots, (l_{n-1},l_n)\}$
- Let $C$, $D$ be clauses (assume there are no common variables since we can rename vars). $K$ is a U-resolvent of $C$, $D$ iff there are non-empty $\underline{C}'\subseteq C$, $\underline{D}'\subseteq D$ s.t. σ is a unifier for $\underline{C}'\cup\underline{D}'^-$ and $K=(C\backslash\underline{C}' \cup D\backslash\underline{D}')\sigma$. Note $|\underline{C}'|$, $|\underline{D}'|$ can be >1

$C = \{\neg R(x), R(f(x))\}\ D = \{\neg R(f(f(x))), P(x)\}$ corresponds to

$\langle \forall x\ (\neg R(x) \vee R(f(x)))\ \wedge\ (\neg R(f(f(x))) \vee P(x))\rangle$ equivalent to

$\langle \forall x\ \neg R(x) \vee R(f(x))\rangle\ \wedge\ \langle \forall x\ \neg R(f(f(x))) \vee P(x)\rangle$ equivalent to

$\langle \forall x\ \neg R(x) \vee R(f(x))\rangle\ \wedge\ \langle \forall y\ \neg R(f(f(y))) \vee P(y)\rangle$ corresponds to

$C = \{\neg R(x), R(f(x))\}\ D = \{\neg R(f(f(y)) \vee P(y)\}$

so I will rename variables in clauses as I see fit

Recall from the Prenex Normal Form algorithm (let $z$,$y$ be $x$ in the example)

$\langle \forall x :: \phi\rangle \wedge \langle \forall y :: \psi\rangle\ \equiv\ \langle \forall z :: \phi\frac{z}{x} \wedge \psi\frac{z}{y}\rangle$ **where $z$ is not free in LHS**

# U-resolvent example

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$$C = \{\neg R(x), R(f(x))\} \quad D = \{\neg R(f(f(x))), P(x)\}$$

$$\{\neg R(x), \underline{R(f(x))}\} \quad \{\underline{\neg R(f(f(y)))}, P(y)\}$$

$$\sigma = f(y) \leftarrow x$$

$$\{\neg R(f(y)), P(y)\}$$

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- Try this: $C = \{\neg S(c,x), \neg S(x,x)\}$, $D = \{S(x,x), S(c,x)\}$

One possible U-resolution step

$$\{\underline{\neg S(c,x)}, \neg S(x,x)\} \qquad \{\underline{S(c,y)}, S(y,y)\}$$

$$\sigma = x \leftarrow y$$

$$\{\neg S(x,x), S(x,x)\}$$

Tautology, so useless

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$$\sigma = c \leftarrow y, x$$

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$$\{\neg S(c,x), S(c,x)\}$$

All are tautologies (useless)

# U-resolvent example

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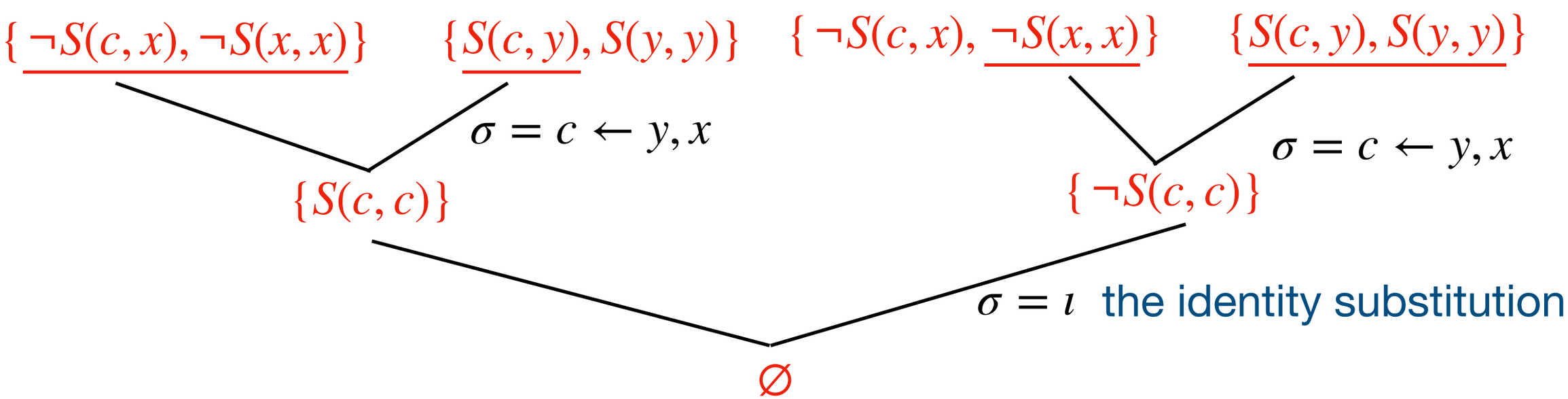

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$$\{\underline{\neg S(c,x), \neg S(x,x)}\} \qquad \{\underline{S(c,y), S(y,y)}\}$$

$$\sigma = c \leftarrow y, x$$

$$\varnothing$$

- This is the Barber of Seville problem: Prove that there is no barber who shaves all those, and those only, who do not shave themselves.

$$\neg\langle \exists b\ \langle \forall x\ S(b,x) \equiv \neg S(x,x)\rangle\rangle$$

# U-resolvent example

- Try this: $C = \{\neg S(c,x), \neg S(x,x)\}$, $D = \{S(x,x), S(c,x)\}$

$$\{\underline{\neg S(c,x), \neg S(x,x)}\} \qquad \{\underline{S(c,y), S(y,y)}\}$$

$$\sigma = c \leftarrow y, x$$

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- This is the Barber of Seville problem: Prove that there is no barber who shaves all those, and those only, who do not shave themselves.

$\neg\langle \exists b\ \langle \forall x\ S(b,x) \equiv \neg S(x,x)\rangle\rangle$ *is valid*

= *{FOL}* $\langle \exists b\ \langle \forall x\ S(b,x) \equiv \neg S(x,x)\rangle\rangle$ *is UNSAT*

= *{Skolem}* $\langle \forall x\ S(c,x) \equiv \neg S(x,x)\rangle$ *is UNSAT*

= *{Clause Form}* $\{\{\neg S(c,x), \neg S(x,x)\}, \{S(x,x), S(c,x)\}\}$ *is UNSAT*

- And then U-resolution, as above

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- Let $C$, $D$ be clauses (assume there are no common variables since we can rename vars). $K$ is a U-resolvent of $C$, $D$ iff there are non-empty $\underline{C'}\subseteq C$, $\underline{D'}\subseteq D$ s.t. $\sigma$ is a unifier for $\underline{C'}\cup\underline{D'}^-$ and $K=(C\backslash\underline{C'} \cup D\backslash\underline{D'})\sigma$. Note $|\underline{C'}|$, $|\underline{D'}|$ can be >1
- Lemma: Let $C$, $D$ be clauses. Then
  - every resolvent of ground instances of $C$, $D$ is a ground instance of a U-resolvent of $C$, $D$
  - every ground instance of a U-resolvent of $C$, $D$ is a resolvent of ground instances of $C$, $D$
- Let $\mathcal{K}$ be a set of ground clauses, $\mathrm{Res}(\mathcal{K})=\mathcal{K}\cup\{K \mid K \text{ is a resolvent of } C,D\in\mathcal{K}\}$
- Let $\mathcal{K}$ be a set of FO clauses, $\mathrm{URes}(\mathcal{K})=\mathcal{K}\cup\{K \mid K \text{ is a U-resolvent of } C,D\in\mathcal{K}\}$
- Let $\mathrm{Res}_0(\mathcal{K})=\mathcal{K}$, $\mathrm{Res}_{n+1}(\mathcal{K})=\mathrm{Res}(\mathrm{Res}_n(\mathcal{K}))$, $\mathrm{Res}_\omega(\mathcal{K})=\cup_{n\in\omega}\mathrm{Res}_n(\mathcal{K})$
- Let $\mathrm{URes}_0(\mathcal{K})=\mathcal{K}$, $\mathrm{URes}_{n+1}(\mathcal{K})=\mathrm{URes}(\mathrm{URes}_n(\mathcal{K}))$, $\mathrm{URes}_\omega(\mathcal{K})=\cup_{n\in\omega}\mathrm{URes}_n(\mathcal{K})$

# Unification for FOL

- Let $C$, $D$ be clauses (assume there are no common variables since we can rename vars). $K$ is a U-resolvent of $C$, $D$ iff there are non-empty $\underline{C}' \subseteq C$, $\underline{D}' \subseteq D$ s.t. σ is a unifier for $\underline{C}' \cup \underline{D}'^{-}$ and $K = (C \setminus \underline{C}' \cup D \setminus \underline{D}')\sigma$. Note $|\underline{C}'|$, $|\underline{D}'|$ can be >1
- $G(K)$ is the set of ground instances of $K$, $G(\mathcal{K}) = \cup_{K \in \mathcal{K}} G(K)$
- Lemma: $\mathrm{Res}_n(G(\mathcal{K})) = G(\mathrm{URes}_n(\mathcal{K}))$ and $\mathrm{Res}_\omega(G(\mathcal{K})) = G(\mathrm{URes}_\omega(\mathcal{K}))$
- Lemma: $\varnothing \in \mathrm{Res}_\omega(G(\mathcal{K}))$ iff $\varnothing \in \mathrm{URes}_\omega(\mathcal{K})$
- For Φ a set of ∀ formulas in CNF: $G(\mathcal{K}(\Phi)) = \mathcal{K}(G(\Phi))$, where $\mathcal{K}(\Phi)$ is set-representation of CNF
- Theorem: For Φ a set of ∀ formulas in CNF, Φ is Sat iff $\varnothing \notin \mathrm{URes}_\omega(\mathcal{K}(\Phi))$
  - Proof: Φ is Sat iff G(Φ) is (propositionally) Sat iff $\mathcal{K}(G(\Phi))$ is Sat iff $G(\mathcal{K}(\Phi))$ is Sat iff $\varnothing \notin \mathrm{Res}_\omega G(\mathcal{K}(\Phi))$ iff $\varnothing \notin \mathrm{URes}_\omega \mathcal{K}(\Phi)$

# FOL Checking with Unification

- FO validity checker: Given FO $\phi$, negate & Skolemize to get universal $\psi$ s.t. Valid($\phi$) iff UNSAT($\psi$). Let $G$ be the set of ground instances of $\psi$ (possibly infinite, but countable). Let $G_1$, $G_2$ …, be a sequence of finite subsets of $G$ s.t. $\forall g \subseteq G, |g| < \omega$, $\exists n$ s.t. $g \subseteq G_n$. $\exists n$ s.t. Unsat $G_n$ iff Unsat $\psi$ (and Valid $\phi$)
- Unification: intelligently instantiate formulas
- FO validity checker w/ unification: Given FO $\phi$, negate & Skolemize to get universal $\psi$ s.t. Valid($\phi$) iff UNSAT($\psi$). Convert $\psi$ into equivalent CNF $\mathcal{K}$. Then, Unsat $\psi$ iff $\varnothing \in \text{URes}_\omega(\mathcal{K})$ iff $\exists n$ s.t. $\varnothing \in \text{URes}_n(\mathcal{K})$.
- We say that U-resolution is *refutation-compete*: If Unsat($\mathcal{K}$) then there is a proof using U-resolution (*i.e.*, you can derive $\varnothing$), so we have a semi-decision procedure for validity.
- Recall: in the worst case, going from $\psi$ to equivalent CNF $\mathcal{K}$ is exponential.
- So, SOA provers use a generalization of Tseitin to generate equi-satisfiable CNF by generating new predicates. Your book calls this *definitional CNF*.

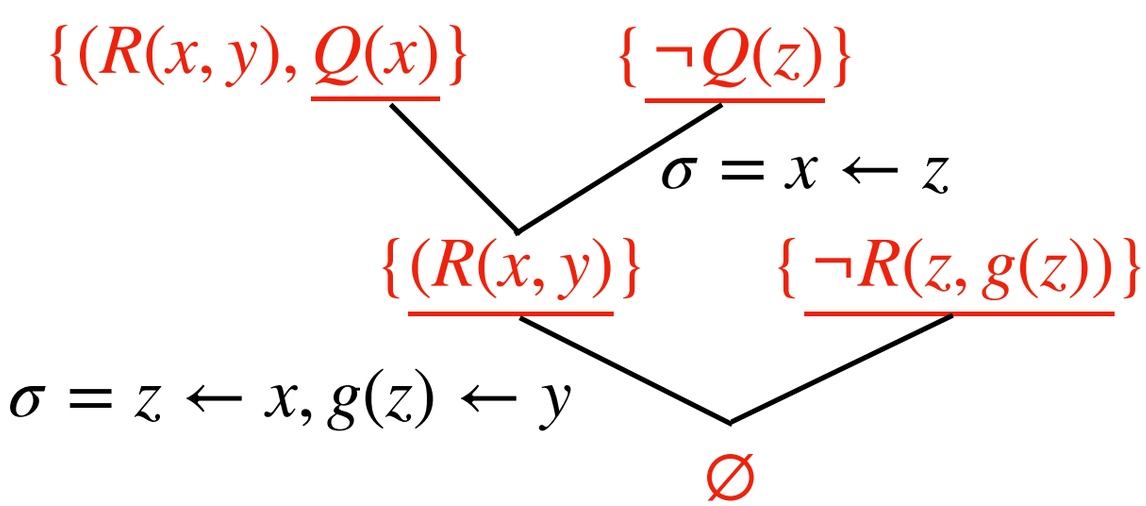
# FOL Checking Examples

- FO validity checker w/ unification: Given FO ϕ, negate & Skolemize to get universal ψ s.t. Valid(ϕ) iff UNSAT(ψ). Convert ψ into equivalent CNF $\mathcal{K}$. Then, Unsat(ψ) iff $\varnothing \in \text{URes}_\omega(\mathcal{K})$ iff $\exists n$ s.t. $\varnothing \in \text{URes}_n(\mathcal{K})$.

$$\phi = \neg\langle \forall x, y\ (R(x,y) \vee Q(x)) \ \wedge\ \neg R(x, g(x)) \ \wedge\ \neg Q(y)\rangle$$

$$\psi = \langle \forall x, y\ (R(x,y) \vee Q(x)) \ \wedge\ \neg R(x, g(x)) \ \wedge\ \neg Q(y)\rangle$$

$$\mathcal{K} = \{\{R(x,y), Q(x)\}, \{\neg R(x, g(x))\}, \{\neg Q(y)\}\}$$




So, Unsat(ψ) and Valid(ϕ)

Let *C, D* be clauses (w/ no common variables). *K* is a U-resolvent of *C*, *D* iff there are non-empty $\underline{C'} \subseteq C$, $\underline{D'} \subseteq D$ s.t. σ is a unifier for $\underline{C'} \cup \underline{D'}^-$ and $K = (C \setminus \underline{C'} \cup D \setminus \underline{D'})\sigma$.

Recall

# Subsumption & Replacement

- Let *C, D* be propositional clauses; $C \leq D$, *C subsumes D* if $C \subseteq D$, therefore $C \Rightarrow D$ and we can remove *D* and subsumed clauses
- Let *C, D* be FO clauses; $C \leq D$, *C subsumes D* if $\exists \sigma$ s.t. $C\sigma \subseteq D$ (matching!), hence $C \Rightarrow D$ and we so can remove D and subsumed clauses
- Theorem: For FO clauses, if $C \leq C'$ and $D \leq D'$ then any U-resolvent of *C*' and *D*' is subsumed by *C, D* or a U-resolvent of *C* and *D*.
- Corollary: If *C* is derivable by U-resolution, then $\exists C'$ derivable by U-resolution s.t. $C' \leq C$ and no clause is subsumed by any of its ancestors
- Corollary: If a U-resolution of a non-tautologous conclusion involves a tautology, $\exists$ a U-resolution proof that does not use any tautologies
- So, we can discard tautologies and subsumed clauses
  - Forward deletion: discard generated clauses that are subsumed by an existing clause
  - Backward replacement: if a generated clause subsumes an existing clause replace the existing clause with the newly generated one

# Positive, Semantic Resolution

- *Positive resolution* (Robinson): Refutation completeness is preserved if we restrict resolution so that one of the clauses contains only positive literals
  - Hint: suppose that there are no positive clauses (all literals are positive), then the problem is SAT if you assign all atoms *false*; if there only positive clauses assign all atoms *true*; see proof in book
- Similarly for U-resolution
  - This cuts down the search space dramatically
  - This plays well with subsumption and replacement
- We could have required negative clauses (instead of positive clauses)
- More generally we have *semantic resolution*: if *S* is an Unsat set of FO clauses and *I* is an interpretation of the symbols used in *S*, there is a U-resolution proof of Unsat(*S*) where each U-resolution step involves a clause that is not true in *I*
  - Positive resolution is a special case where *I* assigns false to all atoms